SCHEDULING: PART 1

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SCHEDULING

Book: Elwood S. Buffa and Rakesh K. Sarin, Modern Production / Operations Management, 8th edition, John Wiley and Sons, 1994



Job Shop Scheduling

- A job shop is a process-focused production system. Production is to order, and a large number of different products are produced, each in relatively small volume.
- Examples of job shops include machining shops, computer centers, printing centers and a paint shop.
- A production manager of a job shop uses the results of scheduling in several aspects of decision making. These results of scheduling are useful in decisions concerning order acceptance, due date specifications, and product mix considerations.
- The manager decides on a daily basis how many jobs and which jobs to release to the shop for processing. The criteria of machine utilization and customer service will be important.

Job Shop Scheduling

- Consider an industry that produces several different products. However, only one product can be produced at a time.
- Suppose the production manager has to decide on the scheduling of four products. The production times and due dates for which are shown in Table.

Product	Production Time (in days)	Due Date (in days)
A	7	9
B	8	17
C	4	6
D	2	19

- Product A requires 7 days in manufacturing and is due to be delivered in 9 days. The manager has several alternatives for scheduling the production of these 4 products.
- Manager may produce product A first and then product B, followed by product C and finally product D. Else, product D can be produced first, next product B, then product A, and finally product C.

Job Shop Scheduling

- There are 4×3×2×1=24 distinct ways of scheduling the production of these 4 different products.
- The decision facing the production manager is which one of these possible 24 schedules should be chosen?

Product	Production Time (in days)	Due Date (in days)
A	7	9
B	8	17
C	4	6
D	2	19

- This example illustrates the problem of scheduling on a single processor (or a single machine scheduling).
- The analysis of a single processor illustrates many important problems that arise in more complex scheduling situations

Some Definitions

- Each job in the single processor scheduling model is described by two parameters:

  p_i = Processing time for job *i*

  d_i = Due date for job *i*

- In some cases, r_i (the ready time or release time or arrival time) of job *i* may be useful. In the models discussed further, all jobs are available for processing at time zero. Hence, $r_i = 0$ for all jobs.

- The processing time p_i includes set-up time for job *i*. If job *i* is defined as a lot of several identical pieces, then p_i will denote the time required to process the complete lot.

- Due date d_i may be set by customer requirements or by internal planning considerations.

- Variables used in scheduling decisions are mentioned in Table

Symbol	Description
W_i	Waiting time for job *i* (the amount of time job *i* has to wait before the beginning of processing time)
C_i	Completion time of job *i* (the time at which the processing of job *i* is completed)
F_i	Flow time of job *i* (the amount of time a job spends in the system. Actually, $F_i = C_i - r_i$. However, $r_i = 0$. Hence, $F_i = C_i$)
L_i	Lateness of job *i* (amount of time by which the completion time of job *i* exceeds its due date. Mathematically, $L_i = C_i - d_i$. L_i can be either positive or negative)
T_i	Tardiness of job *i* (If L_i is positive, it is called **tardiness**. It is a violation of due date)
E_i	Earliness of job *i* (If L_i is negative, it is called **earliness**. It represents the completion of a job before its due date)

General Job Shop Scheduling Rules

- **Earliest Due Date First (EDD):** Priority is given to the job with the earliest due date (Job C will be chosen for scheduling on machine A at time zero).
- **First In System First Served (FISFS):** Priority is given to the job that arrived in the shop (not on the machine) first.

Product	Production Time (in days)	Due Date (in days)
A	7	9
B	8	17
C	4	6
D	2	19

- **First Come First Served (FCFS):** Priority is given to the processing of the job that arrived at the machine first.

General Job Shop Scheduling Rules

- **Least Slack First (LSF):** Priority is given to the processing of the job that has least slack. Slack is the difference between the due date and the work remaining on the job.
- **Shortest Processing Time (SPT):** Priority is given to the job with the shortest processing time on the machine under consideration.
- **Least Work Remaining (LWR):** Priority is given to the job with the least amount of total processing remaining to be done.
- **Longest Processing Time (LPT):** Jobs with longest processing time are processed first.

Performance Criteria

- The choice of which priority rule to choose depends in part on how each rule performs on different criteria:
  - The priority rules try to minimize completion time, maximize facility utilization, minimize number of jobs in the system, and minimize job lateness.
- These performance criteria incorporate the concept of ***flow time***. It measures the time each job spends waiting plus time being processed.
- For example, if Job C waits 5 days for Job B to be processed and then takes 3 more days of operation time itself, its flow time is 5 + 3 = 8 days

Performance Criteria

- Performance criteria are measured as

$$\text{Average completion time} = \frac{\text{Sum of total flow time}}{\text{Number of jobs}}$$

$$\text{Utilization metric} = \frac{\text{Total job work (processing) time}}{\text{Sum of total flow time}}$$

$$\text{Average number of jobs in the system} = \frac{\text{Sum of total flow time}}{\text{Total job work (processing) time}}$$

$$\text{Average job lateness} = \frac{\text{Total late days}}{\text{Number of jobs}}$$

Performance Criteria

- Computing the lateness of a particular job involves assumptions about the start time during the day and the timing of delivering a completed job.
  - Job lateness = Max {0, yester date + flow time – due date} (A)
- Eq. (A) assumes that today is a work day, work has not yet begun today, and a job finished by the end of a day can be delivered to the customer that same day.
- For example, suppose that today is day 10 (yesterday was day 9). Job A is due tomorrow (day 11) and has a flow time of 1 day. That job would be considered to be completed on time, not late: Max{0, 9 + 1 – 11} = Max{0, – 1} = 0 days late.
- Job B is due on day 25 and has a flow time of 12 days. The lateness of Job B would be: Max{0, 19 + 12 – 25} = Max{0, 6} = 6 days late.

Priority Rules for Dispatching

- ***Example:*** Four different products are to be produced. The production time and due dates are shown in Table. Determine the sequence of processing as per FCFS, SPT, EDD and LPT rules. The work begins on day 1.

Product	Production Time (in days)	Due Date (in days)
A	7	9
B	8	17
C	4	13
D	2	19

- ***Solution:*** As per FCFS rule, the sequence is shown in Table.

Job Sequence	Production Time (in days)	Flow time	Due Date (in days)	Job Lateness
A	7	7	9	0
B	8	15	17	0
C	4	19	13	6
D	2	21	19	2
	21	62		8

Priority Rules for Dispatching

- ***Solution:*** As per **FCFS rule**, the sequence (A-B-C-D)) is shown in Table.

Job Sequence	Production Time (in days)	Flow time	Due Date (days)	Job lateness
A	7	7	9	0
B	8	15	17	0
C	4	19	13	6
D	2	21	19	2
	21	62		8

$$\text{Average completion time} = \frac{\text{Sum of total flow time}}{\text{Number of jobs}}$$

$$= \frac{62}{4} = 15.5 \text{ days}$$

$$\text{Utilization metric} = \frac{\text{Total job work (processing) time}}{\text{Sum of total flow time}} = \frac{21}{62} = 33.87\%$$

$$\text{Average number of jobs in the system} = \frac{\text{Sum of total flow time}}{\text{Total job work (processing) time}} = \frac{62}{21} = 2.95 \text{ jobs}$$

$$\text{Average job lateness} = \frac{\text{Total late days}}{\text{Number of jobs}} = \frac{8}{4} = 2 \text{ days}$$

Priority Rules

- ***Solution:*** As per **SPT rule**, the sequence (D-C-A-B) is shown in Table.

Job Sequence	Production Time (in days)	Flow time	Due Date (days)	Job lateness
D	2	2	19	0
C	4	6	13	0
A	7	13	9	4
B	8	21	17	4
	21	42		8

$$\text{Average completion time} = \frac{\text{Sum of total flow time}}{\text{Number of jobs}}$$

$$= \frac{42}{4} = 10.2 \text{ days}$$

$$\text{Utilization metric} = \frac{\text{Total job work (processing) time}}{\text{Sum of total flow time}}$$

$$= \frac{21}{42} = 50\%$$

$$\text{Average number of jobs in the system} = \frac{\text{Sum of total flow time}}{\text{Total job work (processing) time}}$$

$$= \frac{42}{21} = 2 \text{ jobs}$$

$$\text{Average job lateness} = \frac{\text{Total late days}}{\text{Number of jobs}} = \frac{8}{4} = 2 \text{ days}$$

Product	Production Time (in days)	Due Date (in days)
A	7	9
B	8	17
C	4	13
D	2	19

Priority Rules

- ***Solution:*** As per **EDD rule**, the sequence (A-C-B-D) is shown in Table.

Job Sequence	Production Time (in days)	Flow time	Due Date (days)	Job lateness
A	7	7	9	0
C	4	11	13	0
B	8	19	17	2
D	2	21	19	2
	21	58		4

$$\text{Average completion time} = \frac{\text{Sum of total flow time}}{\text{Number of jobs}}$$

$$= \frac{58}{4} = 14.5 \text{ days}$$

$$\text{Utilization metric} = \frac{\text{Total job work (processing) time}}{\text{Sum of total flow time}}$$

$$= \frac{21}{58} = 36.21\%$$

$$\text{Average number of jobs in the system} = \frac{\text{Sum of total flow time}}{\text{Total job work (processing) time}}$$

$$= \frac{58}{21} = 2.76 \text{ jobs}$$

$$\text{Average job lateness} = \frac{\text{Total late days}}{\text{Number of jobs}} = \frac{4}{4} = 1 \text{ day}$$

Product	Production Time (in days)	Due Date (in days)
A	7	9
B	8	17
C	4	13
D	2	19

Priority Rules

- ***Solution:*** As per **LPT rule**, the sequence (B-A-C-D) is shown in Table.

Job Sequence	Production Time (in days)	Flow time	Due Date (days)	Job lateness
B	8	8	17	0
A	7	15	9	6
C	4	19	13	6
D	2	21	19	2
	21	63		14

$$\text{Average completion time} = \frac{\text{Sum of total flow time}}{\text{Number of jobs}}$$

$$= \frac{63}{4} = 15.75 \text{ days}$$

$$\text{Utilization metric} = \frac{\text{Total job work (processing) time}}{\text{Sum of total flow time}}$$

$$= \frac{21}{63} = 33.33\%$$

$$\text{Average number of jobs in the system} = \frac{\text{Sum of total flow time}}{\text{Total job work (processing) time}}$$

$$= \frac{63}{21} = 3 \text{ jobs}$$

$$\text{Average job lateness} = \frac{\text{Total late days}}{\text{Number of jobs}} = \frac{14}{4} = 3.5 \text{ days}$$

Product	Production Time (in days)	Due Date (in days)
A	7	9
B	8	17
C	4	13
D	2	19

Result of different priority Rules

Rules	Average completion time (in days	Utilization metric (%)	Average number of jobs in system	Average lateness (days)
FCFS	15.5	33.87	2.95	2
SPT	10.2	50	2	2
EDD	14.5	36.21	2.76	1
LPT	15.75	33.33	3	3.5

- Based on computations, it is found that LPT is least effective.
- SPT is superior in two metrics.
- EDD is superior in average lateness.

FCFS: first come first served

SPT: shortest processing time

EDD: earliest due date

LPT: longest processing time

- **Shortest processing time (SPT)** is the preferable technique for minimizing job flow and minimizing the average number of jobs in the system.
  - Its disadvantage is that long-duration jobs may be continuously pushed back in priority in favor of short-duration jobs.
- **Earliest due date (EDD)** minimizes maximum tardiness, which may be necessary for jobs that have a very heavy penalty after a certain date. EDD works well when lateness is an issue.
- **First come first served (FCFS)** does not score well on most criteria (but neither does it score particularly poorly). It has the advantage, however, of appearing fair to customers, which is important in service systems.